

Binary · AS · Practice

U02a · 24 written + 6 pen-first = 30 marks · Name: _____

Unless a real-paper citation is shown, questions are ours. Values are unsigned. Where a question retains low bits, use exactly that stated model.

1. Ours: write the register

(a) Convert **91** to eight-bit binary. Show your working. [2]

(b) Convert **172** to eight-bit binary. Show your working. [2]

2. Ours: read the register

(a) Give the denary value of **10110011**. Show the place values you add. [2]

(b) Give the denary value of **01111110**. Show the place values you add. [2]

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3. Ours: capacity and the zero label

(a) For an eight-bit register, state the number of patterns and the largest unsigned denary value. [2]

(b) For a 12-bit register, state the number of patterns and the largest unsigned denary value. [2]

4. Ours: the carry that does not fit

(a) Add **10011010** and **01100110**. Show your carries and the full mathematical sum in binary. [2]

(b) This machine retains only the lowest eight bits of the sum. State the retained eight-bit pattern. [1]

(c) Explain why unsigned overflow occurs in this calculation. [1]

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5. Ours: exam-shape rehearsal · controller states

A controller uses a 10-bit unsigned state register. This is a fresh authored instance of the bit-pattern counting question shape.

- (a) State the number of distinct binary patterns the 10-bit register can represent. [1]

6. Ours: a game score with an explicit rule

A score is 250 and the player earns 10 points. The storage model keeps only the lowest eight bits of the mathematical result.

- (a) Show the mathematical total in denary and binary, with positional or addition working. [2]

- (b) State the retained eight-bit pattern and its denary value. [1]

- (c) Explain why the retained value differs from the mathematical total. [1]

7. Ours: grouping and representation

- (a) State why binary can represent a switch with two states. [1]

- (b) State the number of bits in a nibble. [1]

- (c) State the number of bits in a byte. [1]

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8. Ours: six answers make a number · pen first [6]

 **Paper first, then the classroom cards.** No Python file is needed. Six cards are labelled by weights 1, 2, 4, 8, 16, 32. A number appears on a card exactly when that weight is used in its binary representation. Answer before trying the cards. These six marks are human marked.

(a) For the number **27**, write YES or NO in card order **1, 2, 4, 8, 16, 32**. Show the weights you used. [2]

(b) A player answers **NO, YES, NO, YES, NO, YES** in that same order. Write the six-bit pattern (largest weight first) and its denary value. [2]

(c) Add a seventh card. State its weight and the largest number the seven cards can identify when at least one answer is YES. [2]

Keyboard companion: continue Karel 05 LightTrail and 06 TheLongHall, your first loops. This companion is outside the 30 marks.

Above and beyond · outside the total. A counter has three faces per wheel: 0, 1, 2. Predict what follows 022. Explain the denary values of 022 and 100 using place values. Why does a carry happen at a different moment?

Additional space - label the question number.
